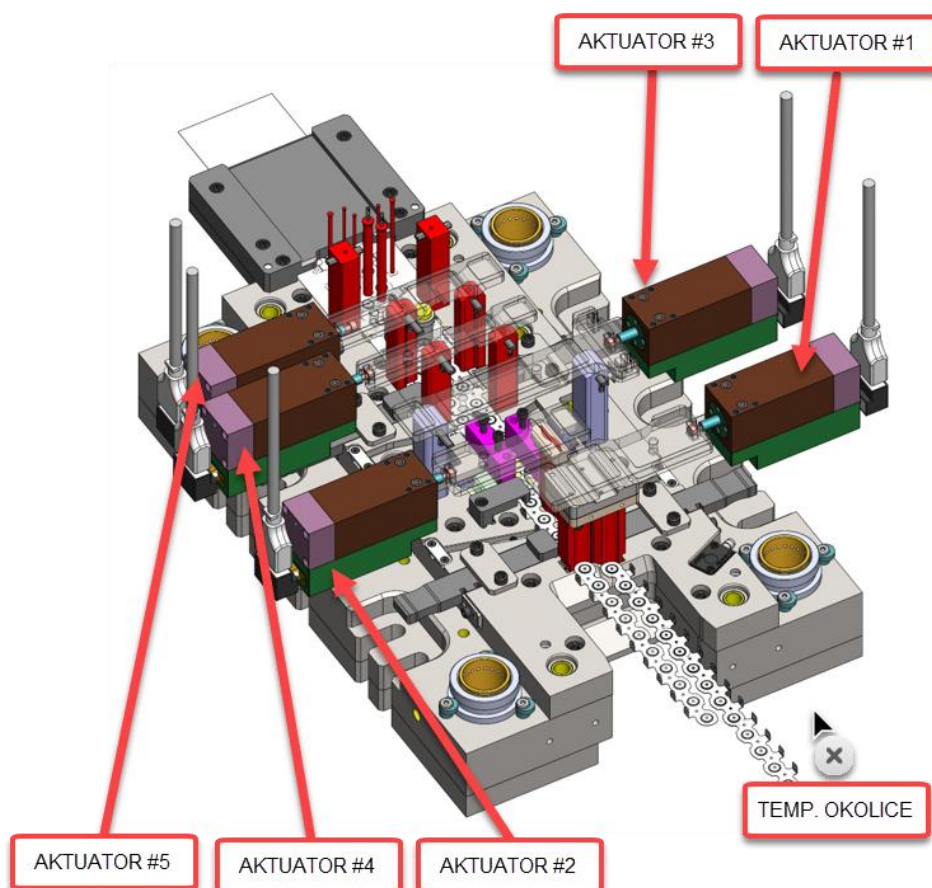


STROKOVNO POROČILO-- razvoj DD realno okolje

Avtor: Dejan Rožič

Datum: 07.05.2025

Projekt: PMO



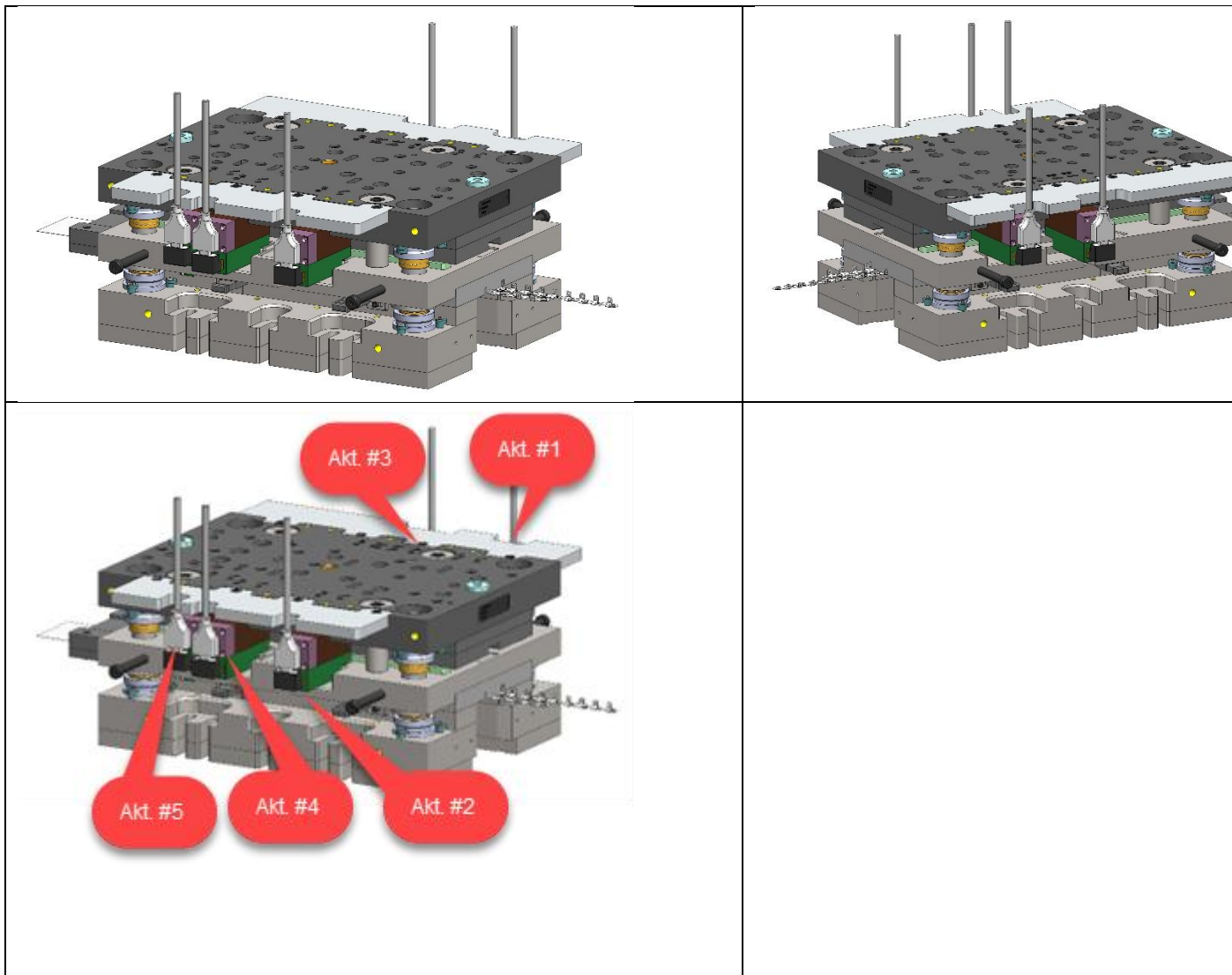


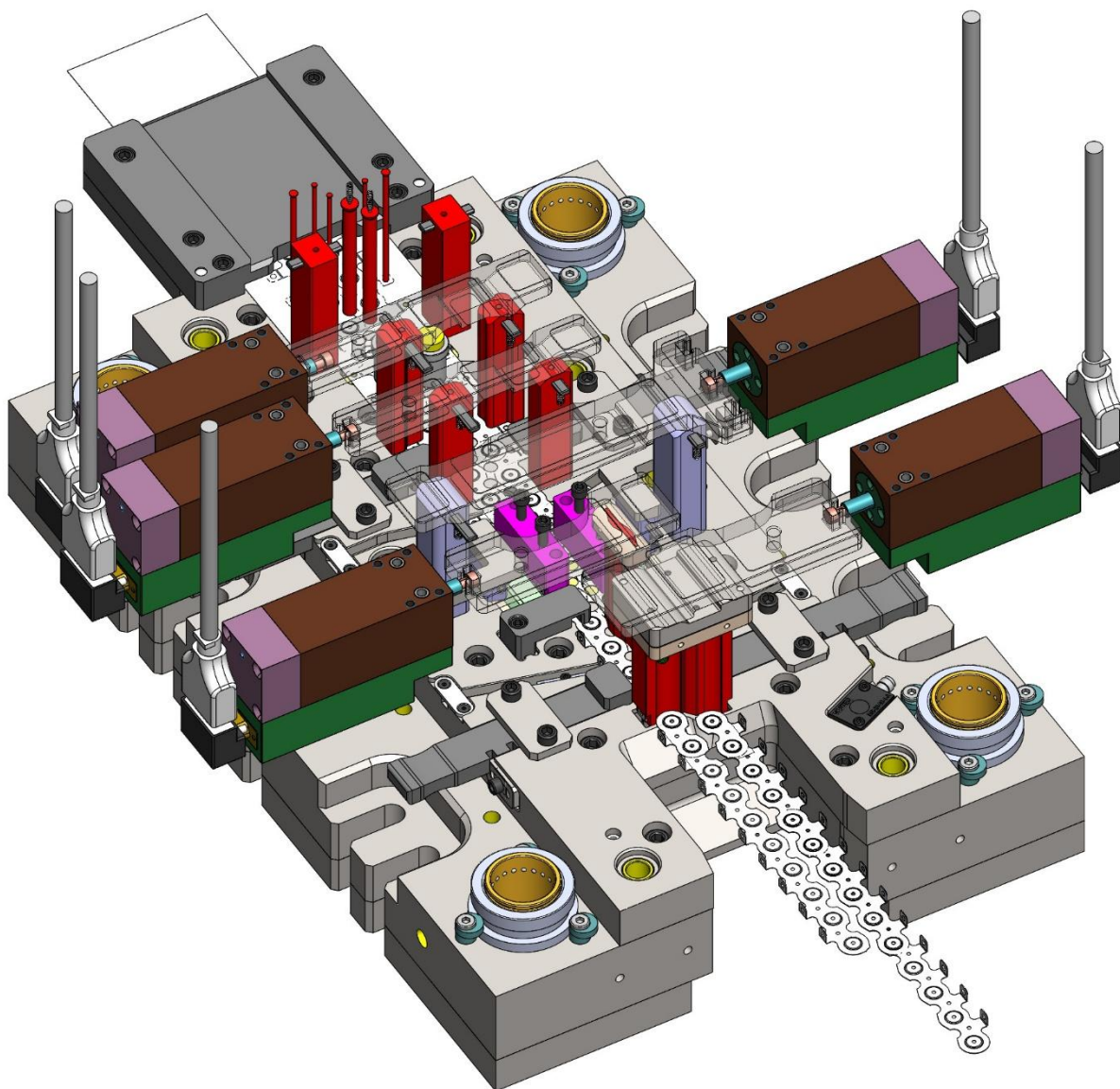
Sensors

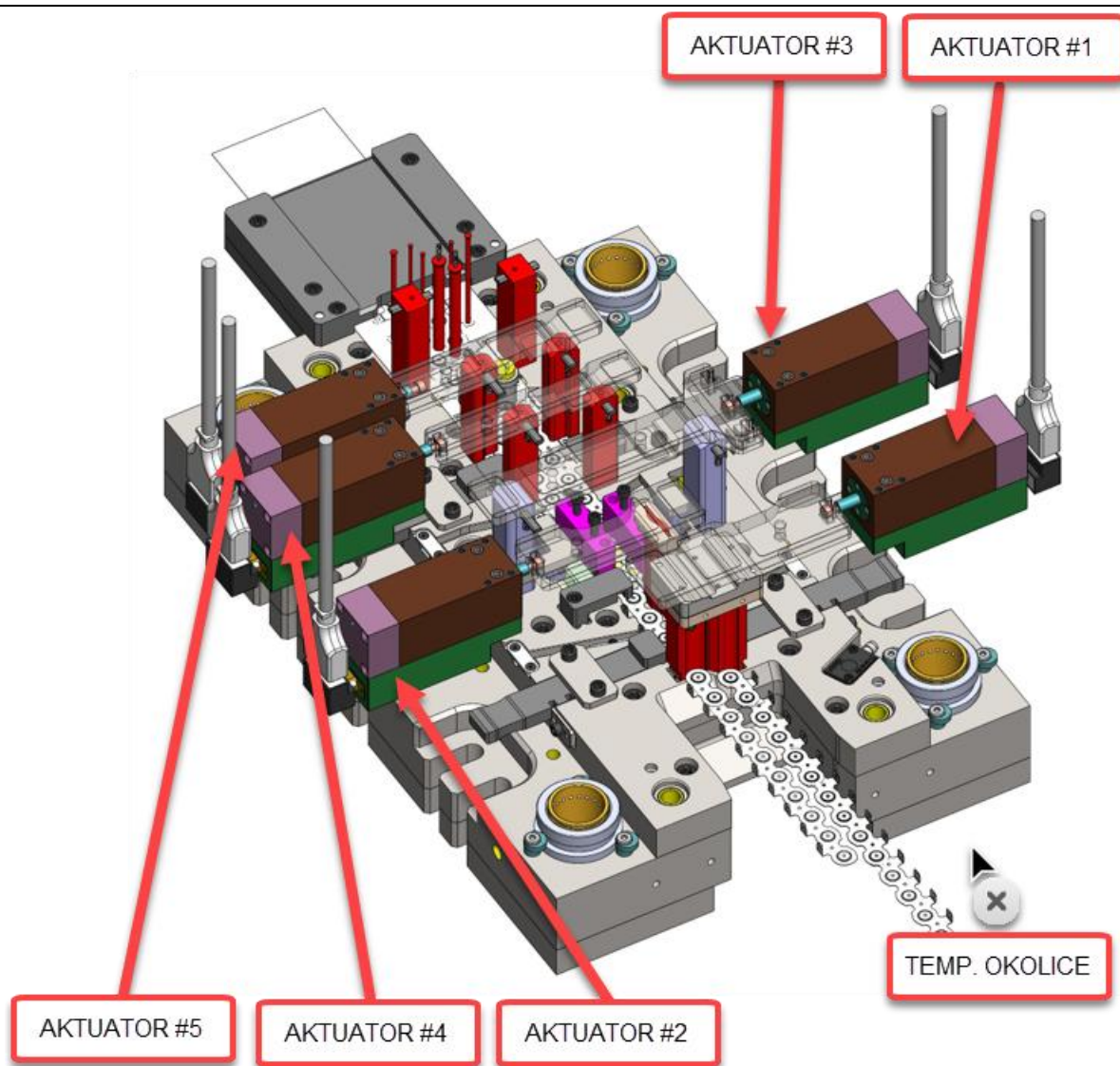
Sensors data for digital twin:

Name : Type	#	Zone	Description
Environment sensor	1	4	Environment temperature
Actuator#1 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator
Actuator#2 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator
Actuator#3 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator
Actuator#4 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator
Actuator#5 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator

*Topics are send at constant interval: 2Hz [can be adjusted]







Sensor limits 50N configuration - temperature range

Sensor	Measuring range	Result
Case sensor	T > 45 °C T > 40°C AND T ≤ 45°C T > 150°C OR T ≤ 0 °C T > 0°C AND T < 150°C	Error: overheating Warning: temperature Not connected Connected range
Current sensor / Force sensor*	I > 7,10A I > 5,39A AND I ≤ 7,10A I > 1,92A AND I ≤ 5,39A I > 4,53A AND I ≤ 5,39A	Error: overrange Warning: current Working range Lubrication required
Tool sensor & Environment sensor	Δt (Tool - Environment) ≥ 5° 5° < Δt (Tool - Environment) > 2° -2° < Δt (Tool - Environment) ≤ 2° Δt < -2°C Δt > 2°C	Error: overheating Warning temp. Working condition Warning environment Warning environment

*Force-Linearization : $F = 11,71 \cdot I - 3,0$ [N]

50N configuration - times to final temperature

Speed	Environment temperature	Egasi temperature	Actuator case	Reaction time to final temperature
100RPM- >150RPM	22.4 -> 23,5	30 -> 35	22.4- >23,8	50 min
150RPM-> 200RPM	24.2 -> 25,2	35 -> 35	24.2 - >25,8	65 min
200RPM- >250RPM	25.2->25.5	35->40	25.2 - >26,9	7 min
250RPM -> 0RPM	27.2->27.6	40->30	29.3- >28.2	20 min
0RPM -> 250RPM	27.7-> 27.5	30->40	28.2- >29.3	19 min

TOPICS SETTINGS

Topic structure

topic: /machine

Main machine status topic

- Topic: */haulick-32t*
- Description: Root topic for Haulick 32T machine status
- Data Format: JSON

example for the topic:

/haulick-32t		
T_env	string	Environment temperature [°C]
is_in_continuous_mode	bool	Machine in continuous mode
is_in_error	bool	Machine has an error
speed	int	Machine speed [RPM]
current_count	int	Machine counter

Example payload:

```
{ "T_env": 23.5, "is_in_continuous_mode": true, "is_in_error": false, "speed": 150, "current_count": 251000 }
```



topic : /machine/tool-nr

Toll-nr machine status topic

- Topic: */haulick-32t/32777/*
- Description: Root topic for tool status
- Data Format: JSON

example for the topic:

<i>/haulick-32t/32777/</i>		
T_tool	int	Tool temperature [°C]

Example payload:

```
{ "T_tool": 23.5 }
```



topic /machine/tool-nr/type-of-sensor_#number

Actuator/motor status topic

- Topics:
 - /haulick-32t/32777/actuator_1,
 - /haulick-32t/32777/actuator_2,
 - /haulick-32t/32777/actuator_3,
 - /haulick-32t/32777/actuator_4
- Description: Root topic for actuator/motor status
- Data Format: JSON

example for the topic:

/haulick-32t/32712/actuator_1		
T	int	actuator temperature [°C]
I _{max}	int	actuator max. current [A]
I _{min}	int	actuator min. current [A]
F _{max}	int	actuator max. force [N]
F _{min}	int	actuator min. force [N]

Example payload:

```
{ "T": 23.5, "Imax":2.5, "Imin":-1A, "Fmax":50N, "Fmin":-10N }
```

Digital twin:

Same topic is for four other actuators (motors) example:

- /haulick-32t/32777/actuator_1 - motor #1
- /haulick-32t/32777/actuator_2 - motor #2
- /haulick-32t/32777/actuator_3 - motor #3
- /haulick-32t/32777/actuator_4 - motor #4

Response from DT (Digital Twin)

Response topics

Response status topic

- Topics:
 - /DT/response/actuator# - example /DT/response/actuator1, /DT/response/actuator2, /DT/response/actuator3, /DT/response/actuator4
- Description: Root topic from digital twin status
- Data Format: JSON

example for the topic:

/DT/response/actuator#	Data type	Comment
req_lubrication	bool	request lubrication for actuator
is_current_error	bool	



is_current_warning	string	error flag warning flag
is_temeprature_error	bool	
is_temperature_warning	bool	error flag warning flag
T_sensor_disconnected	bool	
I_sensor_disconnected	bool	flag for sensor disconnected flag for sensor disconnected

example for the topic

/DT/response/sensors/temperature		
is_overheated	bool	request lubrication
is_in_range	bool	error flag from digital twin
T_sensor_disconnected	bool	error name - comment

Nodered

Node-RED je odprtokodno orodje za **vizualno programiranje**, namenjeno povezovanju naprav, senzorjev, PLC-jev, baz podatkov in spletnih storitev. Deluje po principu **tokov (flows)**, kjer z grafičnimi vozlišči (nodes) definiramo zajem podatkov, njihovo obdelavo in posredovanje naprej (npr. MQTT, HTTP, InfluxDB, HMI).

Kako deluje:

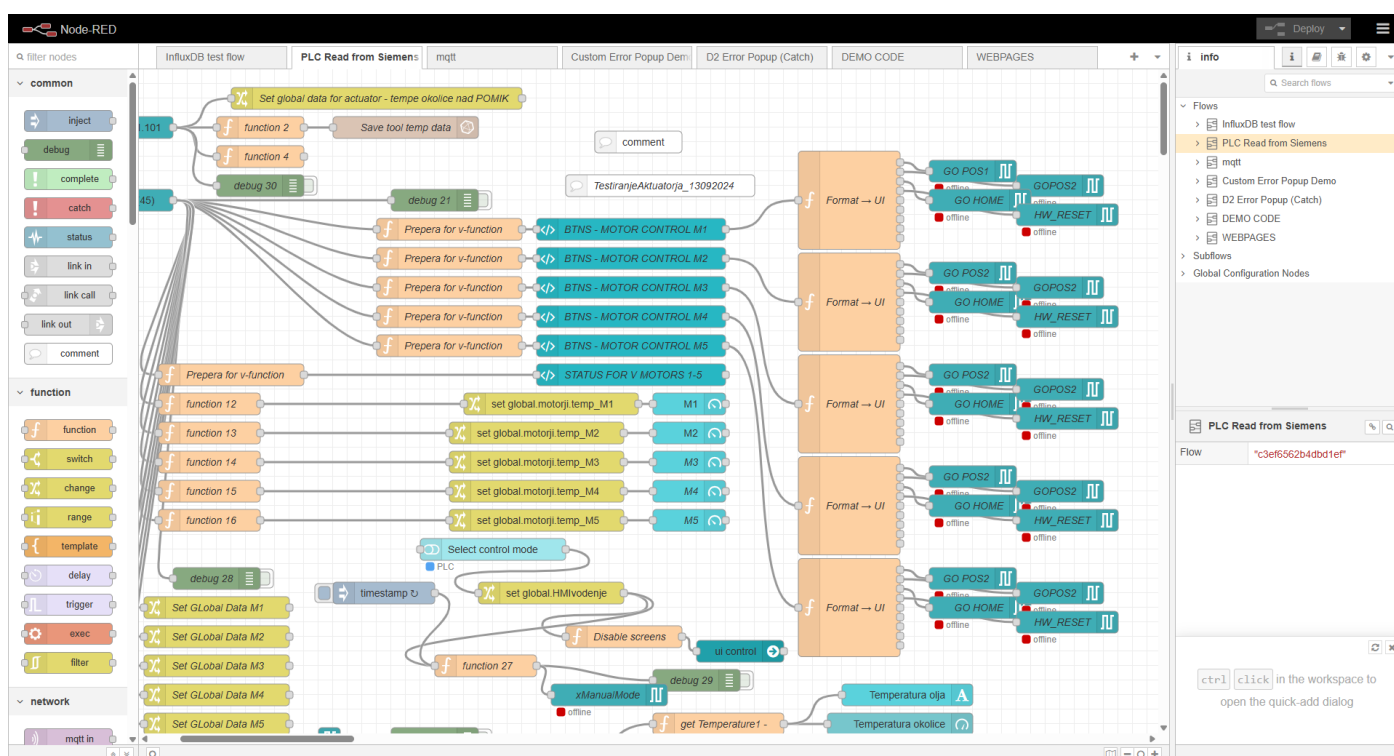
- podatki vstopajo v sistem prek vhodnih vozlišč (PLC, MQTT, API),
- vmes se obdelajo (filtriranje, pretvorbe, logika),
- izhodna vozlišča podatke shranijo, prikažejo ali posredujejo drugim sistemom.

Vloga v RRI razvojnem projektu PMO – Pametno mehatronsko orodje:

Node-RED sem uporabil kot **integracijsko in podatkovno plast** med mehatroničnim sistemom (PLC/senzorji) in IT-okoljem. Omogoča:

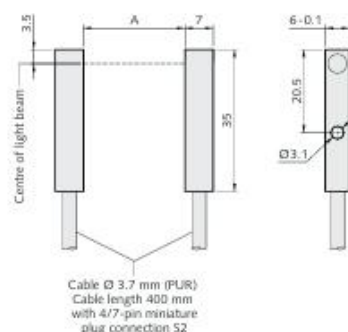
- zbiranje procesnih podatkov (temperatura, napetost, tok, hitrost, stanje stroja),
- pretvorbo in pošiljanje podatkov v **InfluxDB** in druge storitve,
- posredovanje podatkov prek **MQTT** za digitalni dvojček in nadzorne aplikacije,
- hitro prilagodljiv razvoj brez poseganja v osnovno PLC logiko.

S tem Node-RED bistveno prispeva k **hitremu razvoju, modularnosti in razširljivosti** pametnega mehatronskega orodja v okviru RRI projekta.



PG-S2 series

Split infrared light barriers – flat housing F2 with separate amplifier S2



Application

Feed control
Position control
Counting parts

Specifications

Repeating accuracy	< 0.02 mm
Temperature range	0°C – 60°C
EMC according to 2014/30/EU	EN 61000... C
System of protection (for plug-in amplifier types S2-...)	IP 64
General tolerance of the housing	DIN ISO 2768-m

Resolution

Distance A [mm]:	25	90	180
Part diameter [mm]:			
with chopped light amplifier, amplifier level I	> 1.0	> 0.5	–
with chopped light amplifier, amplifier level II	> 1.2	> 0.8	> 0.5
with pulse amplifier	0.1	0.1	0.5

Article

Order No.

Split light barrier, separate transmitter and receiver (without amplifier):

PG-F2-S2	6434-00000-0000
PG-F2-S2 (S)	6434-40000-0000
PG-F2-S2 (E)	6434-50000-0000

Recommendations/instructions for the use of the dual-level chopped light amplifier:

Amplifier level I → High resolution:

Default setting for all split-type versions with small distances between transmitter and receiver, i.e. for $A_{max} < 50\%$!

Amplifier level II → High operational reliability:

Default setting for all split-type versions when used in oil mist surroundings or with large distances between transmitter and receiver ($A_{max} > 50\%$).

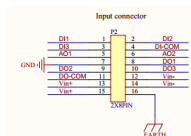
The operational reliability is twice the level of amplifier level I.

Accessories (more accessories, see last pages)

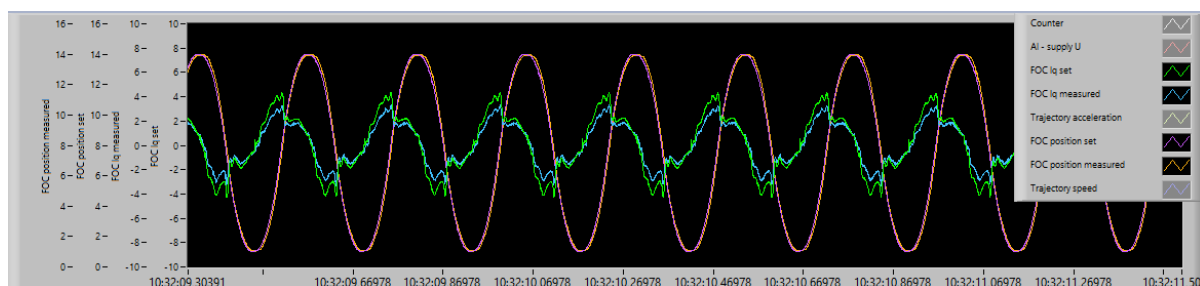
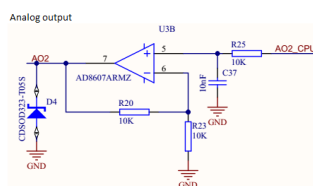
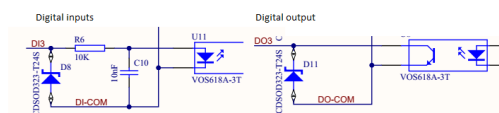
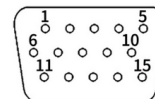
Article

Order No.

Amplifier 4-pin M12:	
S2-W-4-P2	2000-00008-4064
S2-D-4-P	2000-00008-4024
Amplifier 5-pin M16:	
S2-W-5-P2	2000-00008-0064
S2-D-5-P	2000-00008-0024
Mounting block:	
AS2	2300-00000-0002
Extension cable:	
VK-S2/7 (1 m)	2115-00001-0707
VK-S2/4 (1 m)	2115-00001-0404



Input connector - PCB		DSUB15 HD male		Comment	Functionality
PIN	PIN description	PIN	PIN description		
1	DI1	5	Digital input 1	24V input	actuator moves to position 1
2	DI2	4	Digital input 2	24V input	actuator moves to position 2
3	DI3	3	Digital input 3	24V input	rising edge - find home position
4	DI-COM	2	Digital input GND	Common GND for digital inputs	/
5	AO1	1	Analog output 1	0-5V output, 10kHz	settable in parameters - Iq / Id / speed / position / power stage T / actuator T
6	AO2	10	Analog output 2	0-5V output, 10kHz	settable in parameters - Iq / Id / speed / position / power stage T / actuator T
7	AO GND	9	Analog output GND	Common GND for analog outputs	/
8	DO1	8	Digital output 1	open collector output; max 50 mA	actuator reached position 1 (+/- 0.2mm)
9	DO2	7	Digital output 2	open collector output; max 50 mA	actuator reached position 2 (+/- 0.2mm)
10	DO3	6	Digital output 3	open collector output; max 50 mA	homing complete
11	DO-COM	15	Digital output GND	Common GND for digital outputs	/
12	Vin-	14	Supply GND	Power supply - GND	/
13	Vin+	12	Supply 24V	Power supply - 12V - 55V	/
14	Vin-	13	Supply GND	Power supply - GND	/
15	Vin+	11	Supply 24V	Power supply - 12V - 55V	/
16	EARTH (housing)	CONN	connector housing	shield	/



Iq vrednosti so -10A - 10A --> skalirano x0.25 + offset 2.5

Iq = -10A --> AO1 = 0V

Iq = 0A --> AO1 = 2.5V

Iq = 10A --> AO1 = 5V

Actuator T gre od -40°C - 150°C; negativnih temperatur ne bo in bom dal skaliranje 0.04, offset = 0

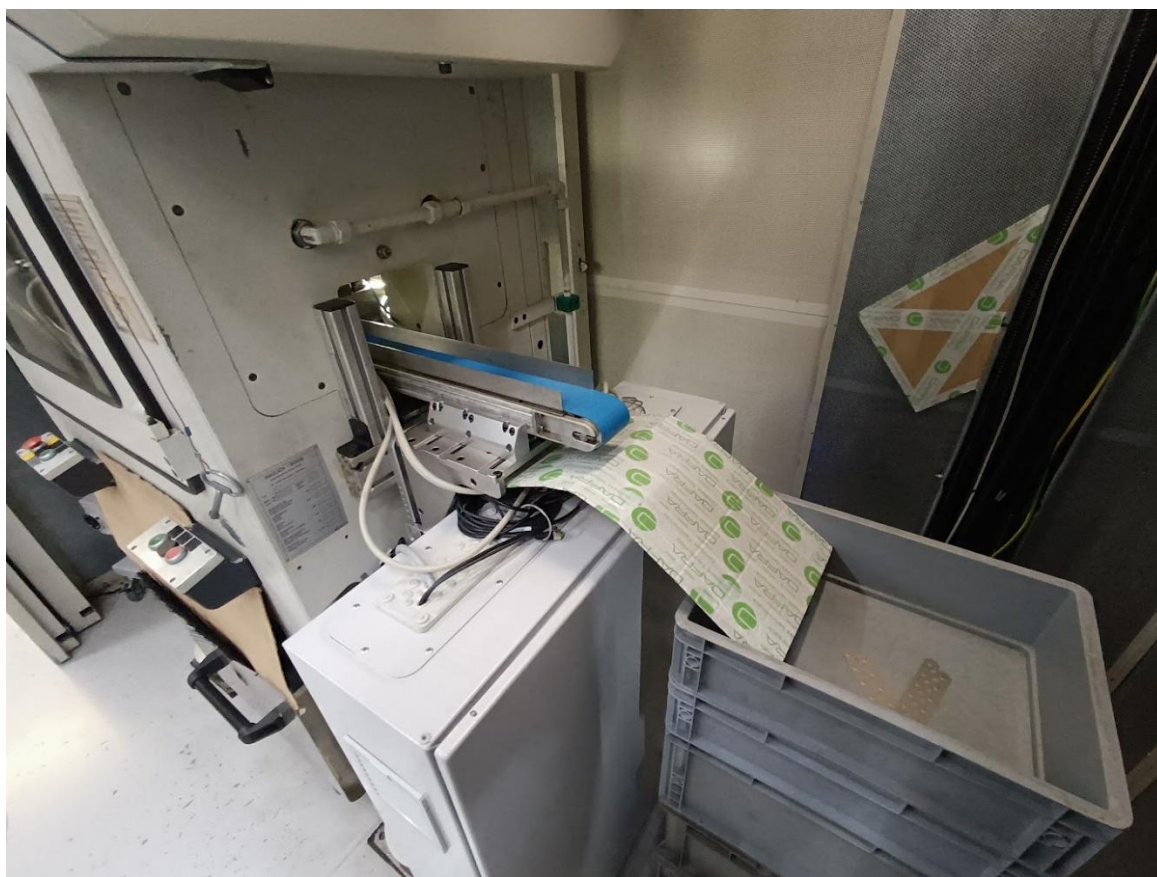
0°C --> AO2 = 0V

50°C --> AO2 = 2V

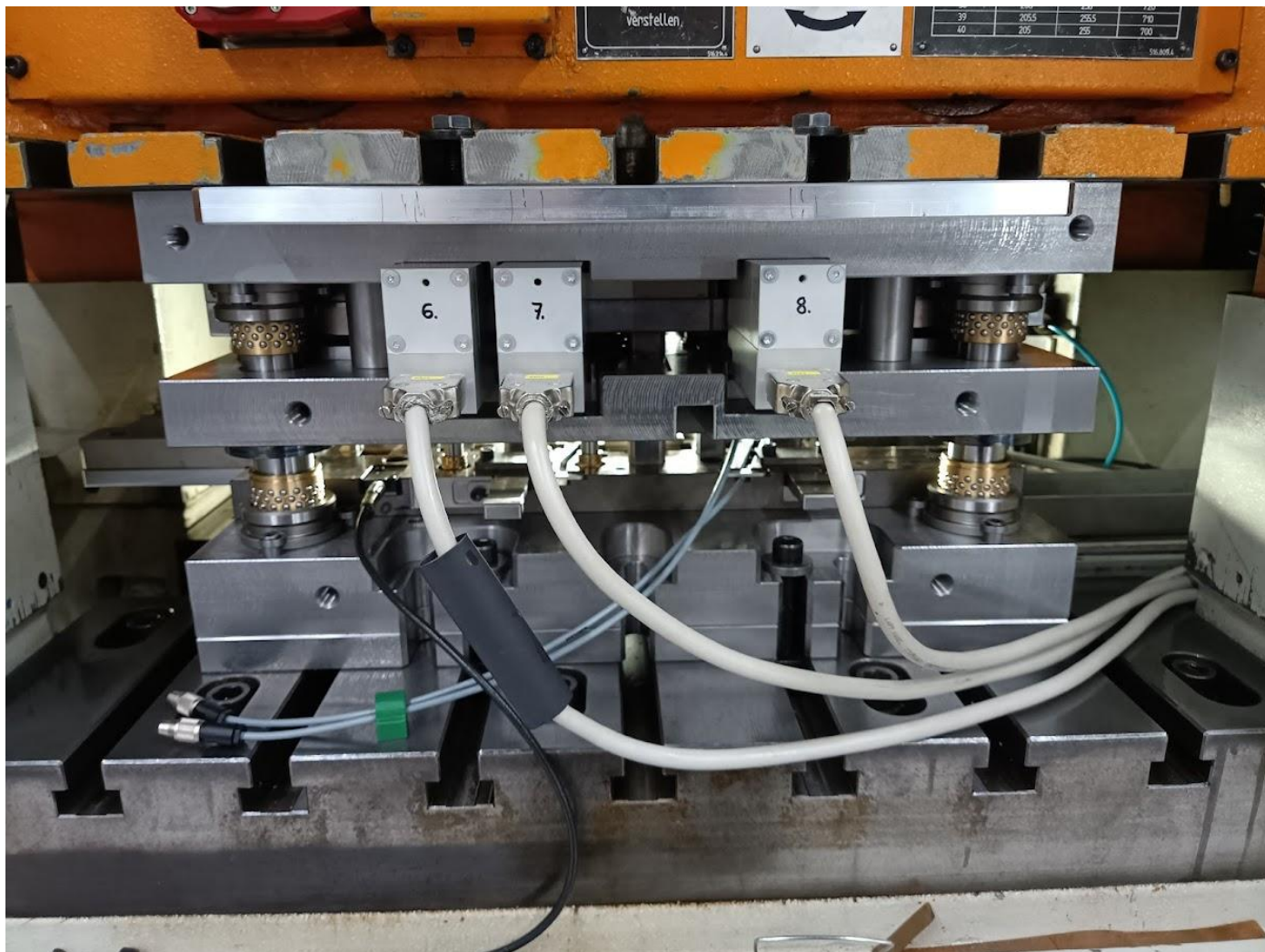
100°C --> AO2 = 4V

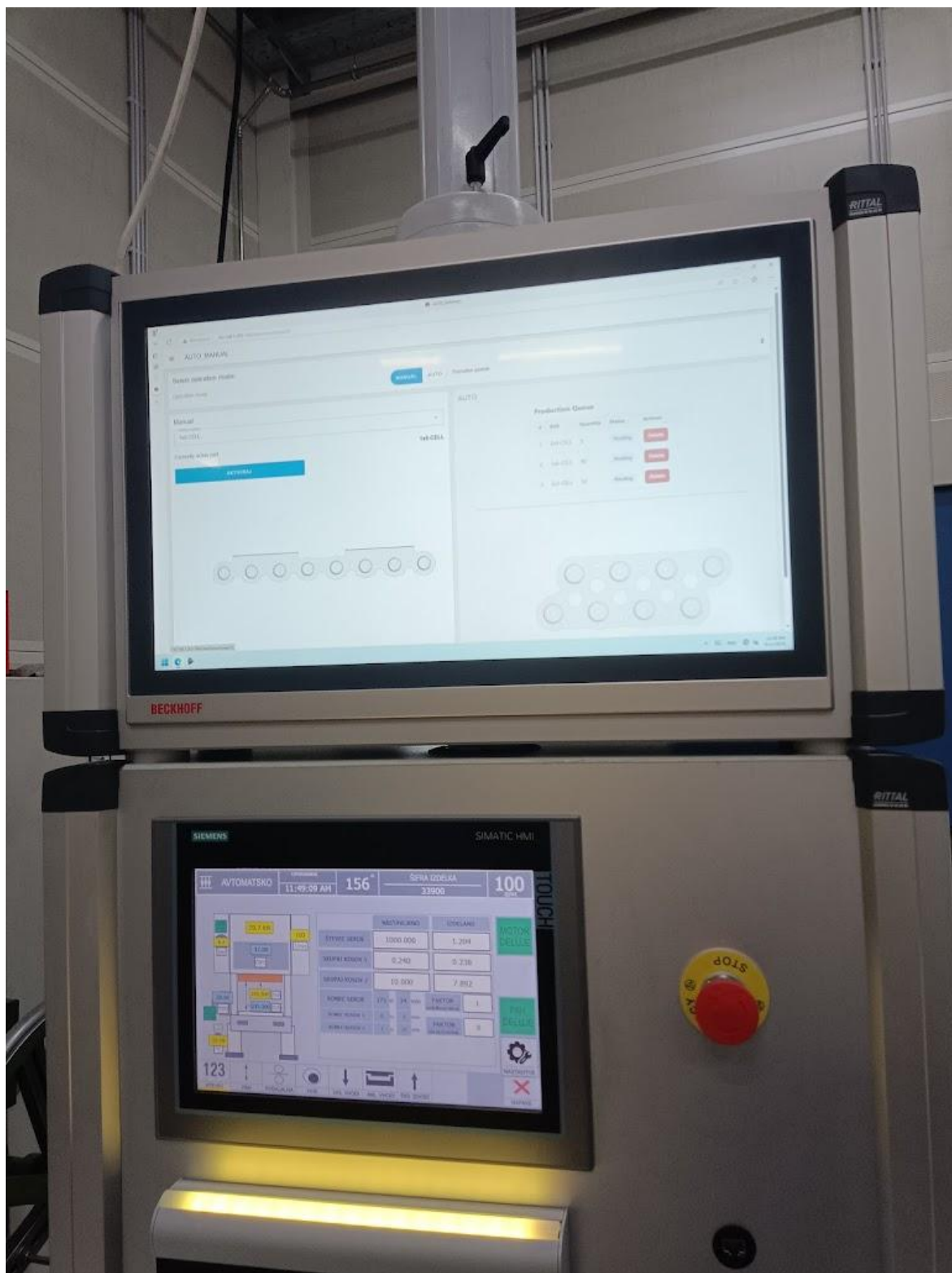


Testiranje digitalnega dvojčka

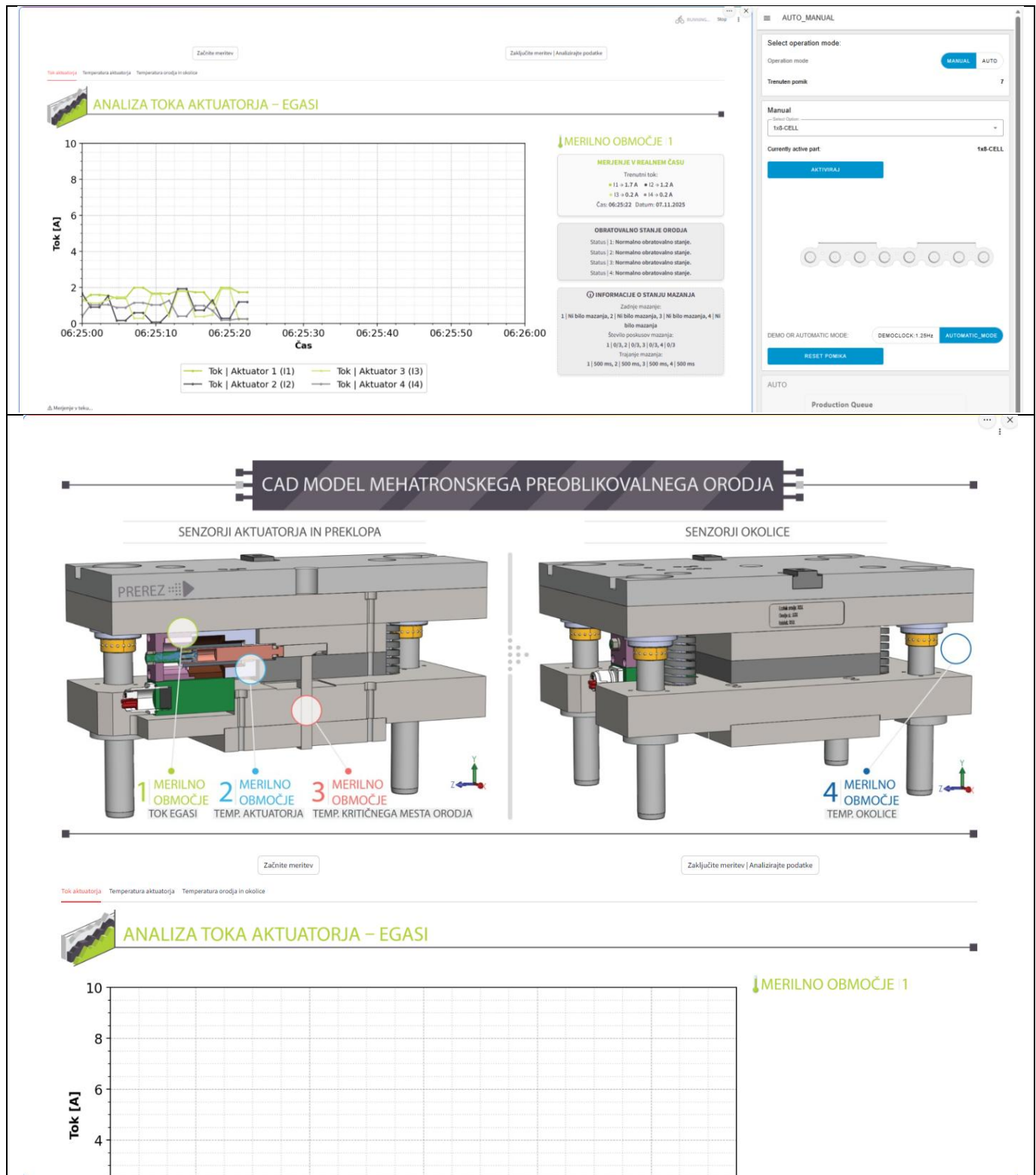




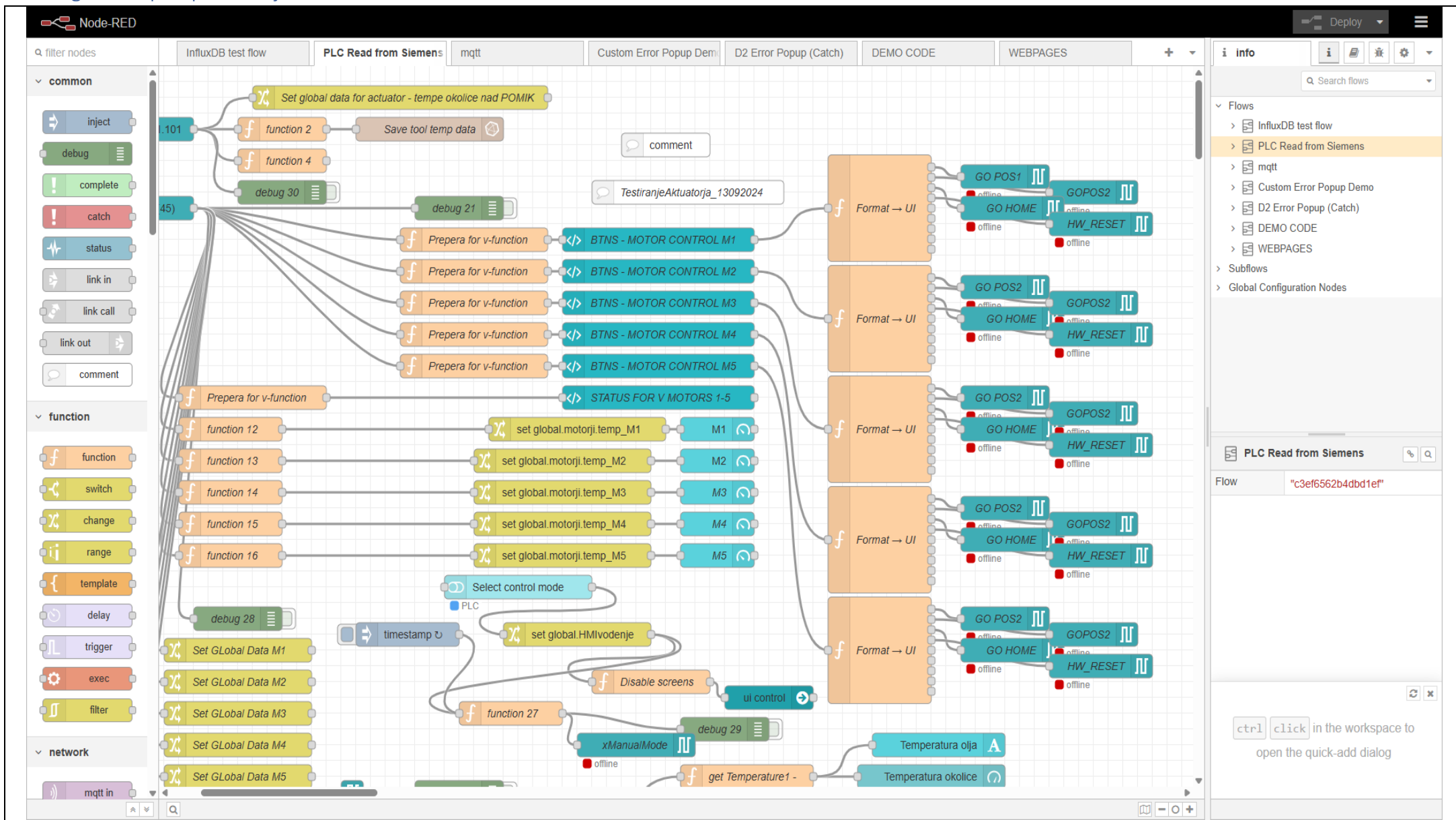


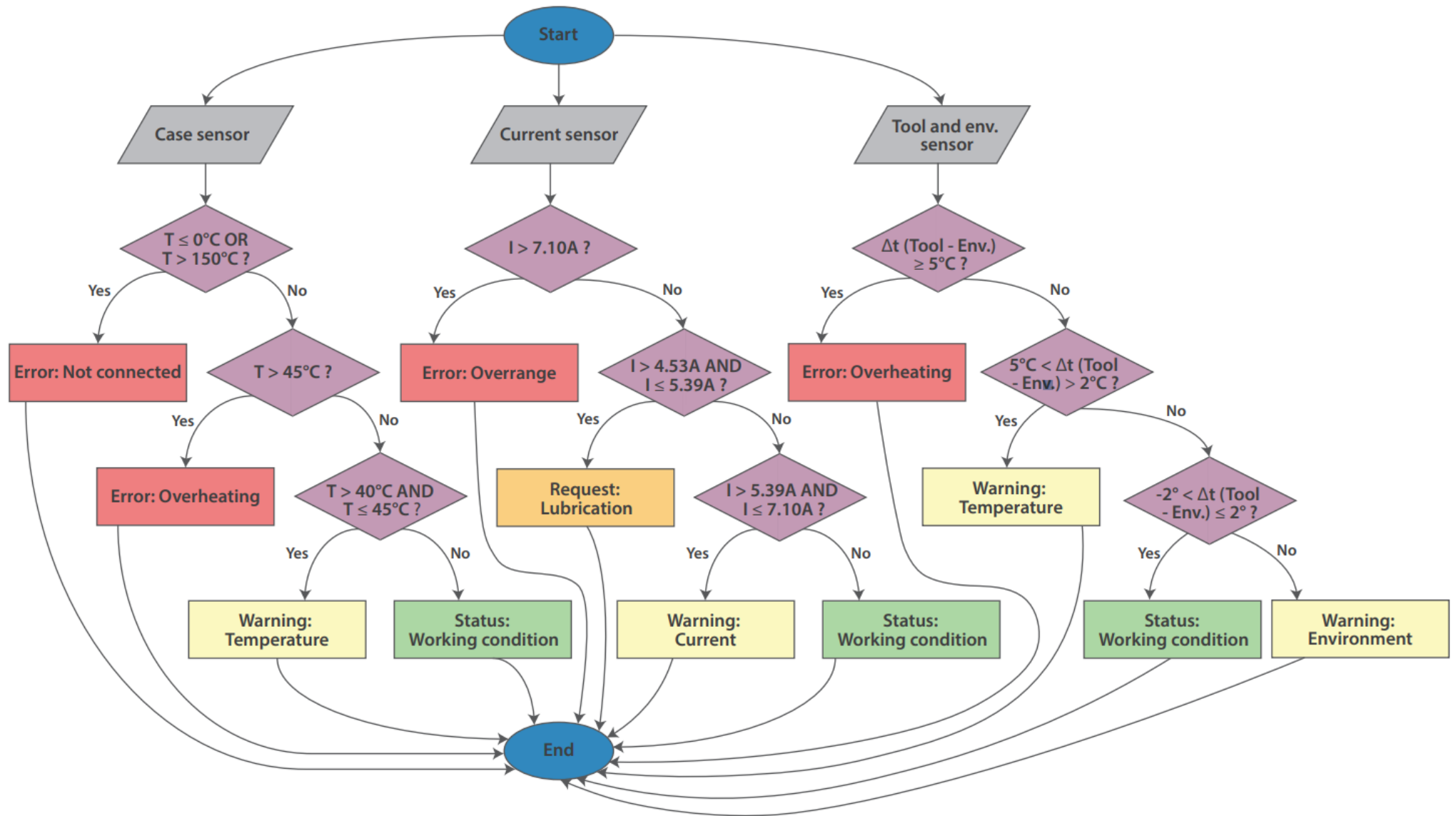


Prikaz zaslona iz aplikacije digitalnega dvojčka

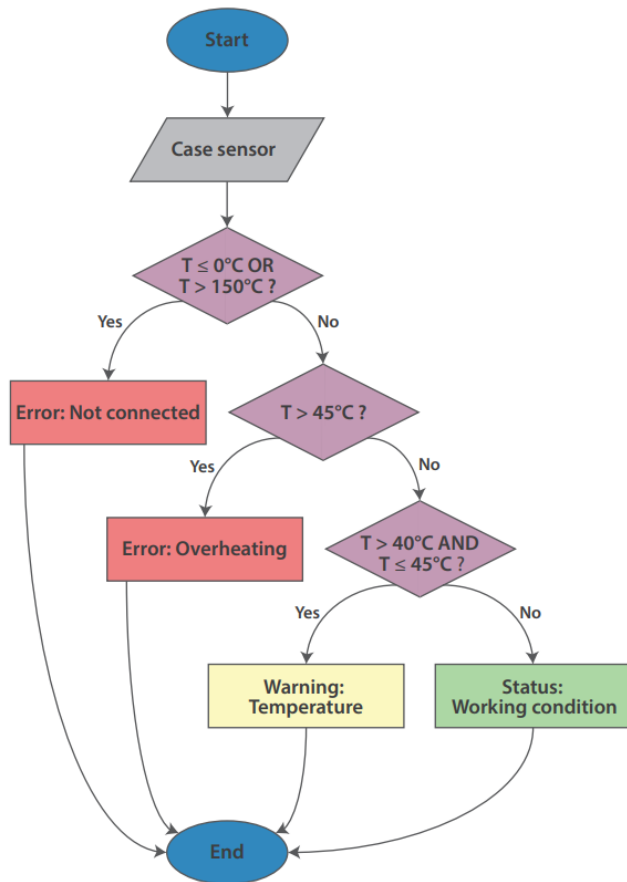


DD – algoritem po optimizaciji

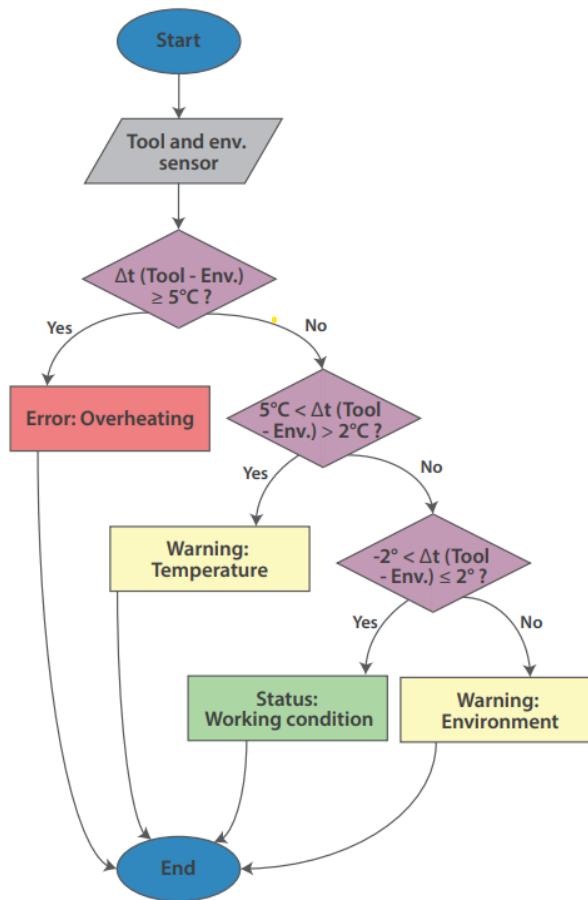




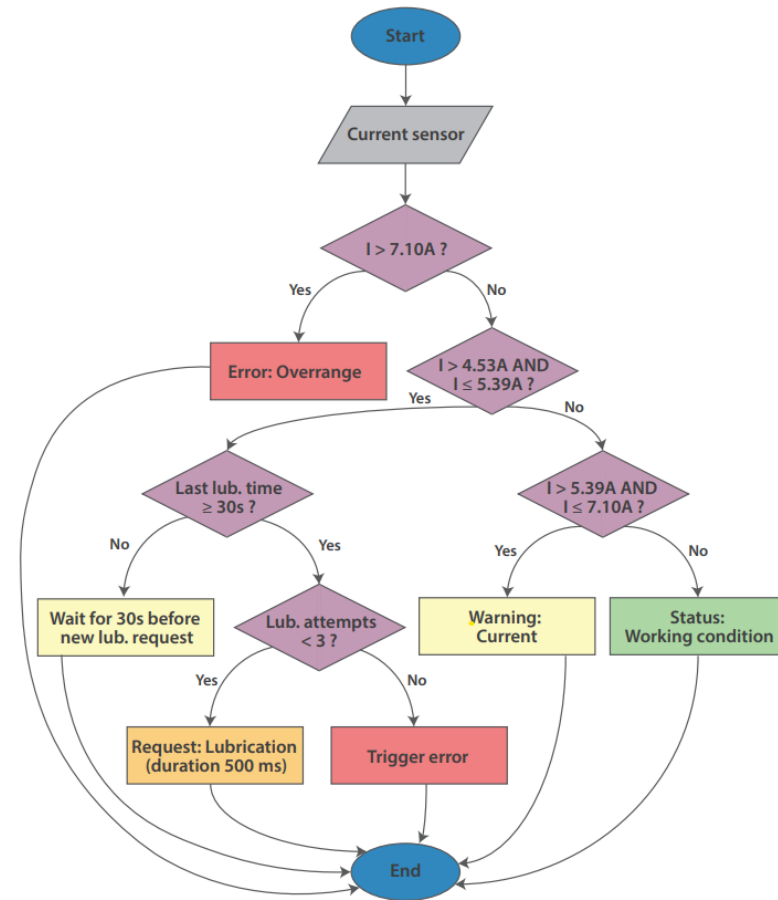
Case sensor



Env. Sensor



Current sensor



☒ VKLOPI/IZKLOPI PRIKAZ OKNA ZA OPTIMIZACIJO OBREMENTITVE AKTUATORJA



OPTIMIZACIJA OBREMENTITVE AKTUATORJA

VNESITE VHODNE PARAMETRE
ZA IZRAČUN SILE



KOT PREKLOPA (°)	25,00	-	+	TRENJE - PESTIČ / PLOŠČA (/)	0,04	-	+
RADIJ PREKLOPA (mm)	0,50	-	+	HITROST ORODJA (min ⁻¹)	100,00	-	+
TRENJE - SKLOP (/)	0,06	-	+	SKUPNA SILA VZMETI (N)	20,00	-	+

IZVEDITE ANALIZO

REZULTAT
IZRAČUNANE SILE

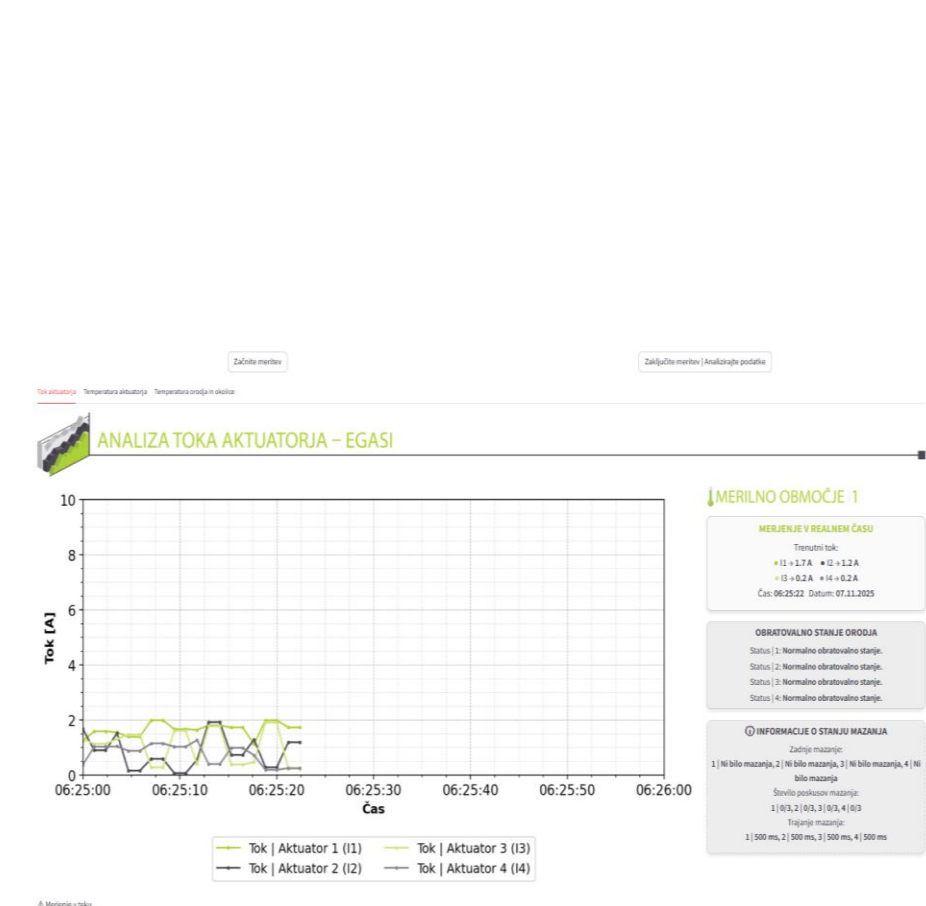
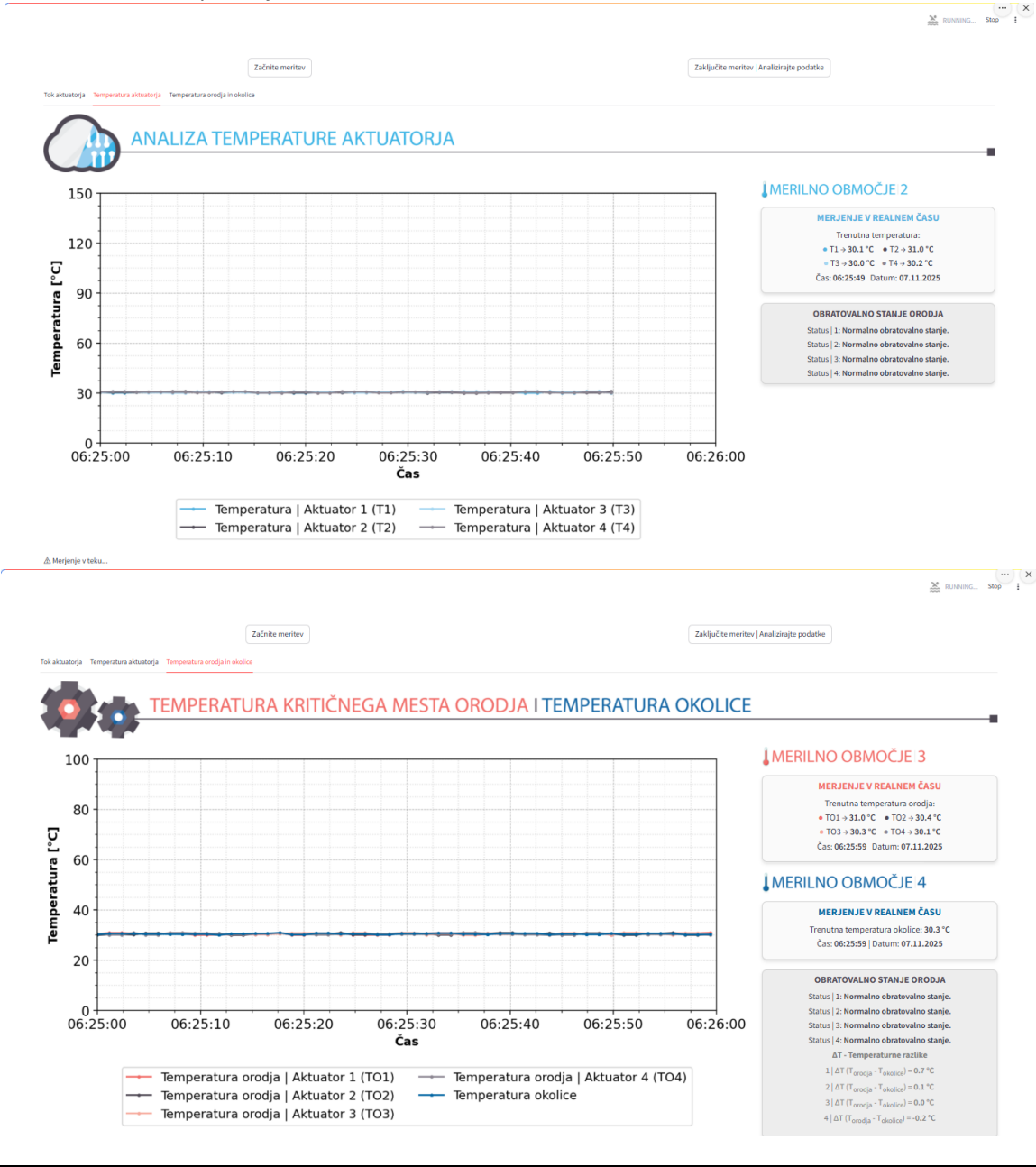


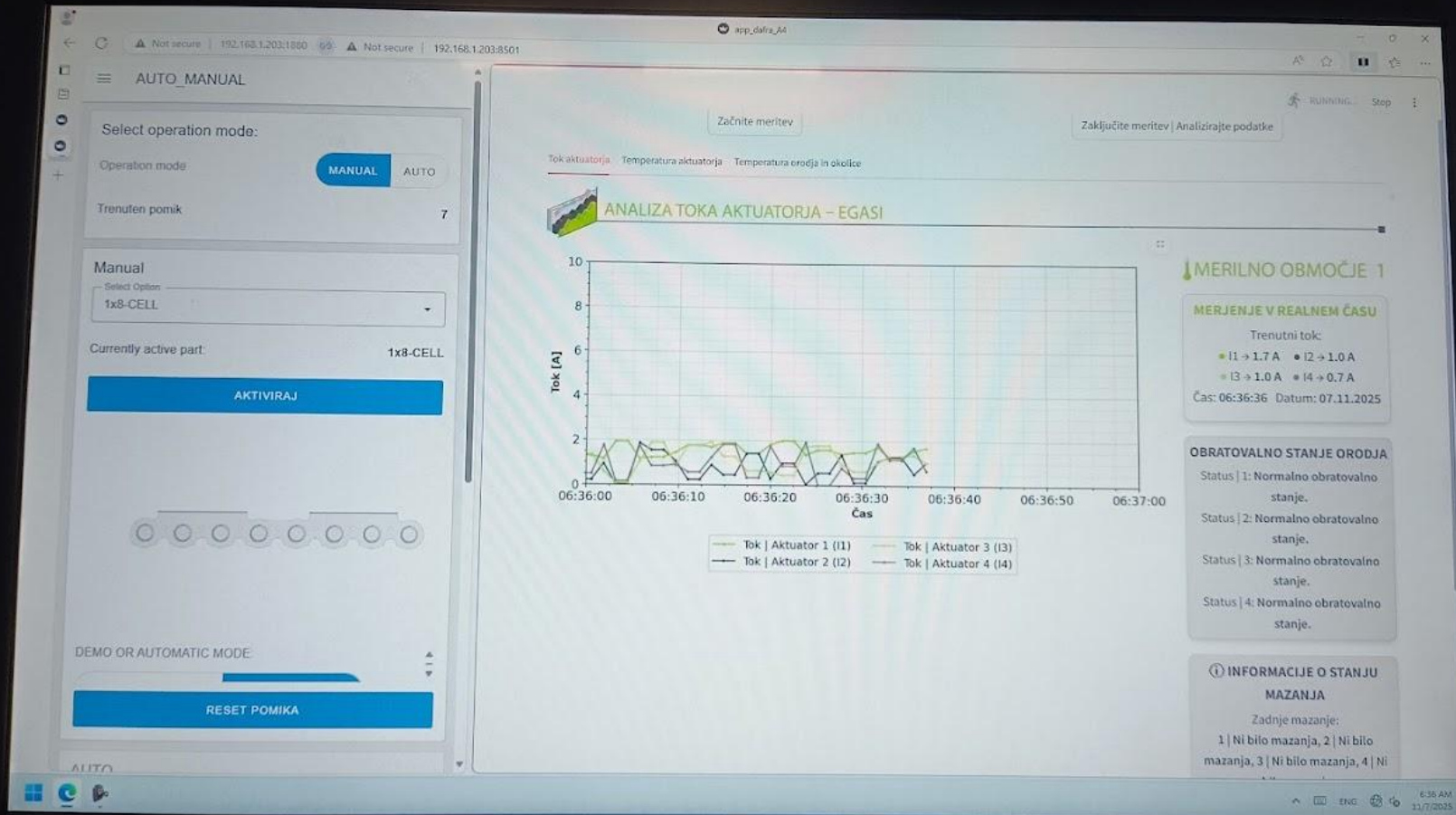
F1 = 2.70 N
F2 = -7.37 N
F3 = -5.62 N

KLIKNITE ZA PODROBNO RAZLAGO IZRAČUNANIH SIL

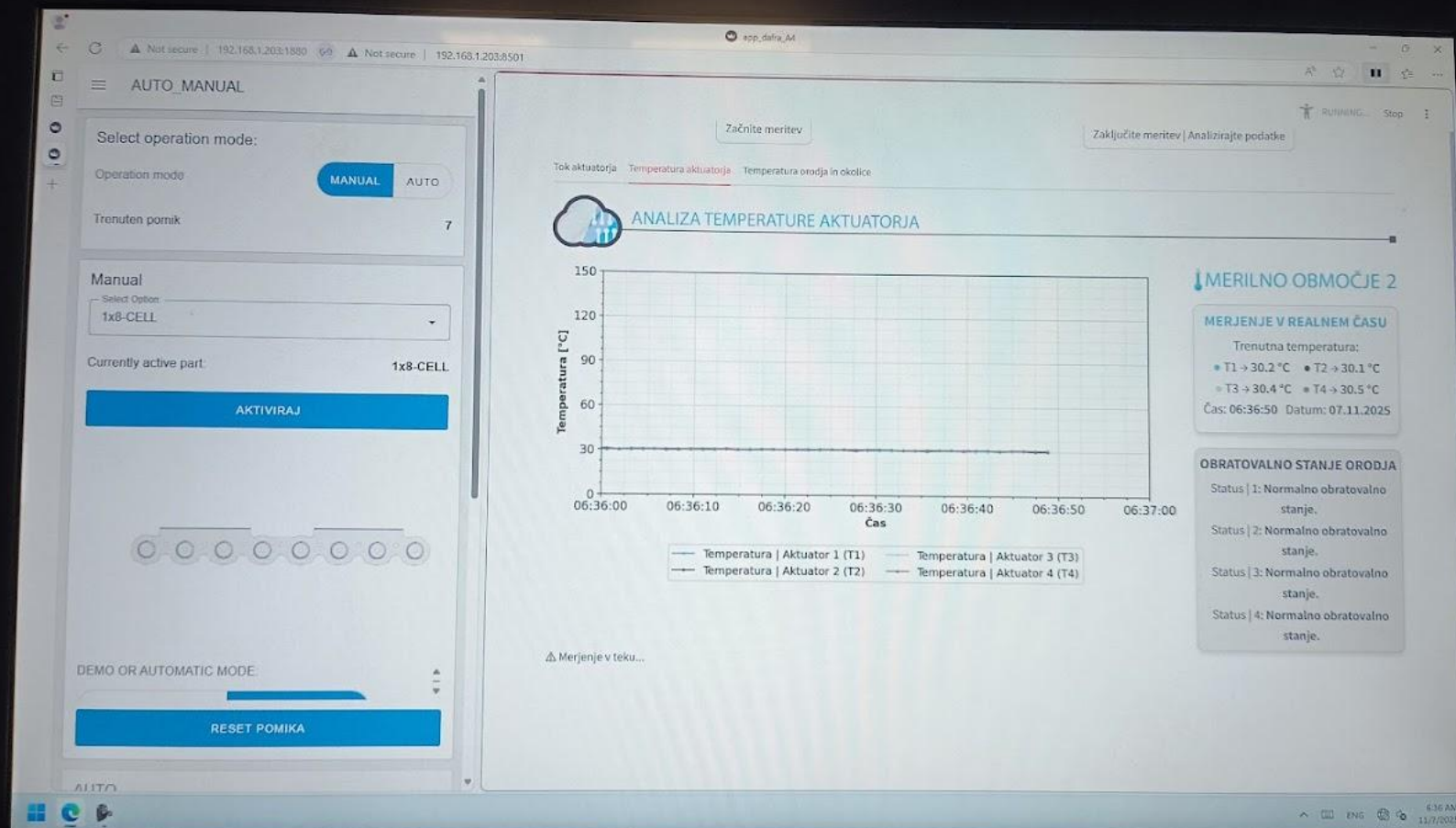
F1 -> Sila trenja zaradi pritiskanja pestiča ob prekop aktuatorja ob pomikanju aktuatorja.
F2 -> Sila zaradi trenja pritiskanja pestiča ob prekop aktuatorja in drsenja prekopnega mehanizma iz obremenjene v sproščeno lego.
F3 -> Sila zaradi trenja pritiskanja pestiča ob prekop aktuatorja in drsenja prekopnega mehanizma iz sproščene v obremenjeno lego.

Slike iz zaslonske aplikacije





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